- 3D Packing (volume-based)
  - Entities
    - Set of items = I = {i}
    - Set of containers K = {k}
  - Parameters
    - Costs
      - Packing items
        - Container independent: c_i \forall i \in I
        - Container dependent: c_{i,k} \forall i \in I, k \in K
      - Using containers: p_k \forall \k \in K
    - Demands
      - Demand of item: d_i \forall i \in I
    - Sizes
      - Weight of item: q_i \forall i \in I
      - Volume of item: v_i \forall i \in I, v_i = l_i*w_i*h_i
    - Capacity
      - Weight limit of container: Q_k \forall k \in K
      - Volume limit of container: V_k \forall k \in K, V_k = L_k*W_k*H_k
    - Compatibility
      - Between items - Set of incompatible item pairs: P = \{(i,j)\} for some i, j \in I
      - Between containers and items - Set of incompatible container-item pairs: O = \{(i,k)\} for some i \in I, k \in K
    - Dimensions (optional if volumes are given)
      - Item dimensions (length, width, height): (l_i, w_i, h_i) \forall i \in I
      - Container dimensions (length, width, height): (L_k, W_k, H_k) \forall k \in K
    - Resources
      - Maximum number of available containers: N
    - Loading times (optional)
      - Loading time per item
        - Container independent: t_i \forall i \in I
        - Container dependent: t_{i,k} \forall i \in I, k \in K
      - Maximum loading time
        - Per container: T_k
        - Total: T_{\max}
    - Load balancing
      - Difference in weight between containers is within a tolerance: \epsilon_Q
      - Difference in volume between containers is within a tolerance: \epsilon_V
  - Decision variables
    - Item placement
      - Number of item i placed in container k: x_{i,k} \in \{0,1,2, ...\} \forall i \in I, k \in K
    - Indicator of item placement (required only if P is defined)
      - Item i is placed in container k (1) and otherwise (0): z_{i,k} \in \{0, 1\} \forall i \in I, k \in K
    - Container usage
      - Container k is used (1) or not (0): y_k \in \{0,1\} \forall k \in K
  - Objectives
    - Minimisation
      - Minimise total cost
        - Item packing cost is independent of containers: \min\ \sum_{k \in K} (\sum_{i \in I} c_i x_{i,k} + p_k y_k)
        - Item packing cost is dependent of containers: \min\ \sum_{k \in K} (\sum_{i \in I} c_{i,k} x_{i,k} + p_k y_k)
      - Minimise space wastage
        - W.R.T. Weight: \min\ \sum_{k \in K} (Q_k - \sum_{i \in I} q_i x_{i,k})
        - W.R.T. Volume: \min\ \sum_{k \in K} (V_k - \sum_{i \in I} v_i x_{i,k})
      - Minimise cost and space wastage with weights
        - Packing cost independent of containers, W.R.T. Weight wastage: \min\ \alpha [\sum_{k \in K} (\sum_{i \in I} c_i x_{i,k} + p_k y_k)] + \beta [\sum_{k \in K} (Q_k - \sum_{i \in I} q_i x_{i,k})]
        - Packing cost independent of containers, W.R.T. Volume wastage: \min\ \alpha [\sum_{k \in K} (\sum_{i \in I} c_i x_{i,k} + p_k y_k)] + \beta [\sum_{k \in K} (V_k - \sum_{i \in I} v_i x_{i,k})]
        - Packing cost dependent of containers, W.R.T. Weight wastage: \min\ \alpha [\sum_{k \in K} (\sum_{i \in I} c_{i,k} x_{i,k} + p_k y_k)] + \beta [\sum_{k \in K} (Q_k - \sum_{i \in I} q_i x_{i,k})]
        - Packing cost dependent of containers, W.R.T. Volume wastage: \min\ \alpha [\sum_{k \in K} (\sum_{i \in I} c_{i,k} x_{i,k} + p_k y_k)] + \beta [\sum_{k \in K} (V_k - \sum_{i \in I} v_i x_{i,k})]
  - Constraints
    - Demand satisfaction: \sum_{k \in K} x_{i,k} \geq d_i \forall i \in I
    - Capacity constraint
      - Weight limit: \sum_{i \in I} q_i x_{i,k} \leq Q_k \forall k \in K
      - Volume limit: \sum_{i \in I} v_i x_{i,k} \leq V_k \forall k \in K
    - Container availability: \sum_{k \in K} y_k \leq N
    - Load balancing
      - Weight based: \sum_{i \in I} (q_i x_{i,k} - q_i x_{i,m}) \leq \epsilon_Q \forall k \neq m \in K
      - Volume based: \sum_{i \in I} (v_i x_{i,k} - v_i x_{i,m}) \leq \epsilon_Q \forall k \neq m \in K
    - Compatibility
      - Between items: z_{i,k} + z_{j,k} \leq 1 \forall (i,j) in P
      - Between container and items: x_{i,k} = 0 \forall (i,k) in O
    - Loading time constraints (optional)
      - Loading time is independent of containers: \sum_{i \in I} t_{i} x_{i,k} \leq T_k \forall k \in K
      - Loading time is dependent of containers: \sum_{i \in I} t_{i,k} x_{i,k} \leq T_k \forall k \in K
      - Total loading time is bounded
        - Loading time is independent of containers: \sum_{k \in K} \sum_{i \in I} t_{i} x_{i,k} \leq T_{\max}
        - Loading time is dependent of containers: \sum_{k \in K} \sum_{i \in I} t_{i,k} x_{i,k} \leq T_{\max}
    - Binary and integrality constraints (implicit)
      - x_{i,k} \in \{0,1,2,...} \forall i \in I, k \in K
      - y_k \in \{0,1\} \forall k \in K
      - z_{i,k} \in \{0,1\} \forall i \in I, k \in K
    - Linking constraints (implicit)
      - Linking x and y: y_k \geq \sum_{i \in I} x_{i,k} \forall k \in K, sufficiently large M
      - Linking x and z: z_{i,k} M \geq x_{i,k} \forall i \in I, k \in K, sufficiently large M